ALKENE

61. (c) $\underset{sp^2}{CH_3} = \underset{sp}{C} = \underset{sp^2}{CH_2}$
Propadiene

Propadiene has both *sp* and sp^2 -hybridized carbon atoms.

62. (a) $C_2H_4 + Cl_2 \xrightarrow{CCl_4} \underset{\text{Ethelene Chloride}}{Cl - CH_2 - CH_2 - Cl}$

63 (b) Ethylene dibromide

Ethene adds bromine via Electrophilic addition reaction to form 1,2-dibromoethane (ethylene dibromide).

64. (b) Paraffins or alkanes are non-polar compounds. Hence soluble in benzene.

65. (b) $CH_3 - CH = CH_2 + HCl \xrightarrow{Peroxide} CH_3 - \underset{\underset{Cl}{|}}{CH} - CH_3$

Peroxide rule is applicable only to HBr .

66. (d) $CH_3 - CH = CH_2 + HI \xrightarrow{\text{Peroxide}} CH_3 - CH_2 - CH_2I$

67 (c) C_2H_5–CHX–CH_3

Follows Markovnikov's rule.

68 (a) Alc. KOH + heat

Gives elimination (Elimination reaction).

69. (c) $CH_3-\underset{H}{\underset{|}{CH}}-\underset{Cl}{\underset{|}{CH}}-\underset{H}{\underset{|}{CH_2}} \xrightarrow[-HCl]{alc.\,NaOH} CH_3CH=CH-CH_3$

70 (c) Vapour distillation

Ethylene is separated due to its volatility and low solubility in water.

71 (c) Both 1,2 and 1,4 addition product

Due to resonance-stabilized intermediate.

72 (c) One of the steps is endothermic in both the cases

Free radical mechanism is not feasible.

73. (c) $CH_3-CH=CH-CH_2-CH_3$ it decolourizes $KMnO_4$ solution because double bond is present.

74 (d) HCN

X is ethanal (from Wacker oxidation).

Aldehyde shows addition with HCN only.

75 (c) Its bond energy is high

Free radical mechanism is not feasible.

76. (d) All of these

It has conjugation, reacts with HBr, and can polymerize.

77. (a) $CH_2 = CH - CH_2 - C \equiv CH + Br_2 \rightarrow CH_2 = CH - CH_2 - \underset{Br}{\underset{|}{C}} = \underset{Br}{\underset{|}{CH}}$

78. (d) $CH_3COCH_3 + PCl_5 \rightarrow CH_3 - CCl_2 - CH_3 + POCl_3$

79. (a) 1 and 4

Catalytic hydrogenation only converts C=C to C–C without changing the carbon skeleton.

- Compounds (1) and (4) already have the branched framework of 2,2-dimethylpropane (neopentane), so they give its isomer after hydrogenation.
- Compounds (2) and (3) have different carbon skeletons, so they form different alkanes.

80. (b) $3R - CH = CH_2 + \frac{1}{2} B_2H_6 \xrightarrow[\text{Ether}]{\text{Dry}} (R - CH_2CH_2)_3 B$

$$\downarrow OH^-, H_2O_2$$

$$\underset{\text{Alcohol}}{3R - CH_2CH_2OH}$$